

Network fundamentals: -

This module will introduce the core concepts of computer networking. Covering everything from Internet protocol (IP), network topologies TCP and UDP protocols of OSI Transport layer. Don't worry if that sounds complicated, as this module has beginner friendly exercises.

What is networking? - Networking Basics: -

Networks are simple things connected. For example your friendship circle: you are all connected because of similar interests, hobbies, skills and so on.

Networks can be found in all walks of life.

→ A city's public transportation system

→ Infrastructure such as national power grid for electricity.

→ Meeting and greeting your neighbours

→ postal system for sending letters and parcels.

you see networking
in everyday
of life

* In computing Networking is the same idea. Ex: - your phone the reason that you have it to access things. We'll cover how devices communicate with each other and rules that follow.

In computing, network can be found anywhere from 2 devices to billions. These devices include everything from your laptop and phone to security camera, traffic lights and even farming!

Networks are integrated into our everyday life. Gathering data for the weather, delivering electricity to homes or even determining who has right of way at a road. Networks are so embedded in modern day, networking is essential concept to grasp in cybersecurity.

* Take a look from below diagram are examples. Alice, Bob and Tim.

have formed their network, well come on to this bit later on.



Network comes in all shapes and sizes, which is something we'll discuss.

Q) What is the key term for devices that are connected together?

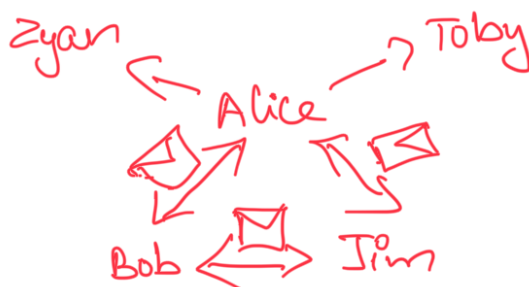
A) Network.

Task 2:— What is the Internet?

Now we learned what a network is how one is defined in computing. (Just devices connected). Let's explore the Internet.

The Internet is one giant network that consists of many, many small networks within itself.

Previous diagram example, Alice made some new friends named Zyan and Toby that she wants to introduce to Bob and Jim. The problem is Alice is only person who speaks the same language as Zyan and Toby. So Alice will have to be the messenger.

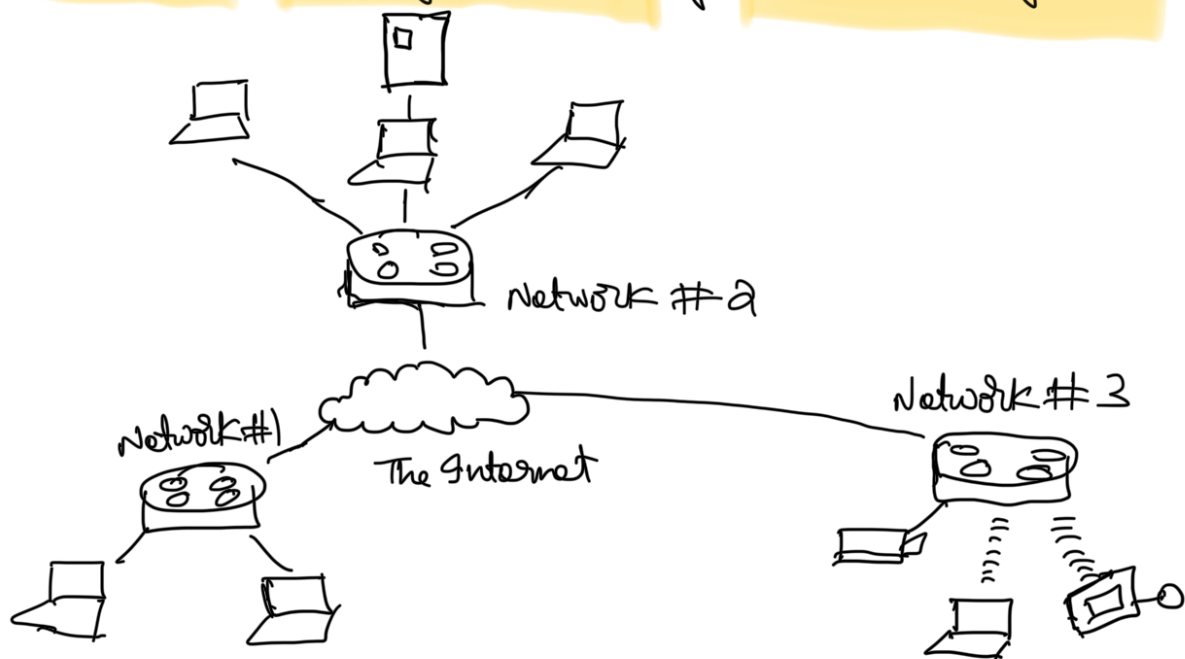


Because Alice can speak both languages, they can communicate to one another through Alice - forming a new network.

The first iteration of Internet was within the ARPANET project in late 1960s. This project was funded by United States defense department and was first documented network in

ation. It wasn't until 1989 when Internet as we know it was invented by Tim Berners-Lee by creation of World Wide Web (WWW). It wasn't until this point that Internet started to be used as repository for storing and sharing information, just like it today. It's just a static website at that time.

Let's relate Alice network of friends to computing devices. The Internet looks much larger version of this sort of diagram.



As previously stated, Internet is made up of many small networks all joined together. These small networks are called private networks, when networks connecting these small networks are called public networks. Or Internet! So to recap network can be one of two types:

1) private network 2) public network.

Q) who invented the world wide web?

A) Tim Berners-Lee

Task 3 will be uploaded on GitHub as well.

Identifying Devices on a Network :-

Devices on a network are very similar to humans, in fact that we have two ways of being identified.

* Our Name * Our fingerprints

Now we can change our name through deedpoll, but we can't however change our fingerprints. Every human has an individual set of fingerprints. Even they change name, there is still an identity.

Even devices have two means of identification, There are:

1) An IP address 2) A media access control (MAC) - think of being similar as serial number.

IP Addresses: —

IP Addresses are set of numbers that are divided into 4 octets.

octet #1	octet #2	octet #3	octet #4
192	168	1	1
0-255	0-255	0-255	0-255

The value of each octet summarise to be

the IP Address of device on network. This number is calculated through a technique known as IP Address and Subnetting, but that is for another day. What's important to understand here is that IP Addresses can change from device to device. but cannot be active simultaneously more than once within the same network.

IP Addresses follow a set of standards known as protocols. These protocols are backbone of networking and for many devices to communicate in same language. we know that devices can be both private and public network. Determine what type of IP address they have: private or public IP Address.

* A public IP address used to identify device on Internet.

* A private IP address is used to identify a device amongst other devices.

Example of private vs public IP Address: —

Device Name	IP Address	IP Address type
Desktop - KJES7FD	192.168.1.77	private
Desktop - KJES7FD	86.157.52.21	public
CMNatic-PC	192.168.1.74	Private
CMNatic-PC	86.157.52.21	public

Desktop - KJES7FD

IP Address: 192.168.1.77 (DHCP)

MAC Address: EC:5C:68:C3:7E:51

IP Address: 192.168.1.74 (DHCP)

CMNatic-PC

MAC Address: 50:3E:AA:E8:3B:64

These two devices use their private IP address to communicate with each other. The private IP addresses are not publicly routable and are only valid through local network. However, these communications are translated through public IP address. This translation is often managed by router using Network Address Translation (NAT). Any data sent to Internet from either of device will be identified by same public IP Address. Public IP Address are given by Internet Service provider (ISP) at a monthly fee (your bill).

My public IP v4 is 86.157.52.21

Location: ENGB

ISP: secure communications TLD